



PSYCH 201B

Statistical Intuitions for Social Scientists

GLM II: Design Matrix

Today's Plan

1. Maths

2. GLM Recap

3. Design Matrix

Maths

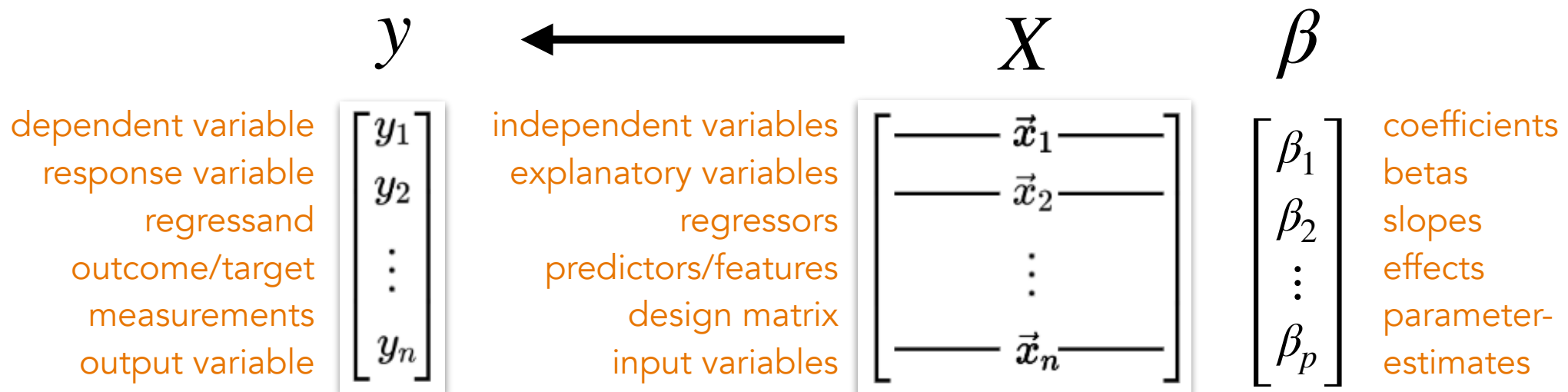
Check out course website!

GLM Recap

The General Linear Model (GLM)

$$y = X\beta + \epsilon$$

Models a **scalar** variable **y** as a linear function of predictor **variables X**



Regression(s) are just “flavors” of GLM

$$y = X\beta + \epsilon$$

Simple (univariate) regression

$$Y_i = \beta_0 + \beta_1 X_{1i} + \epsilon_i$$

one predictor

one column wide

**technically 2 with intercept*

y

dependent variable
response variable
regressand
outcome/target
measurements
output variable

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$



independent variables
explanatory variables
regressors
predictors/features
design matrix
input variables

X

$$\begin{bmatrix} \text{---} \vec{x}_1 \text{---} \\ \text{---} \vec{x}_2 \text{---} \\ \vdots \\ \text{---} \vec{x}_n \text{---} \end{bmatrix}$$

β

$$\begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}$$

coefficients
betas
slopes
effects
parameter-
estimates

Regression(s) are just “flavors” of GLM

$$y = X\beta + \epsilon$$

Multiple regression

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_p X_{pi} + \epsilon_i$$

many predictors

many columns wide

y

dependent variable
response variable
regressand
outcome/target
measurements
output variable

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$



independent variables
explanatory variables
regressors
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X

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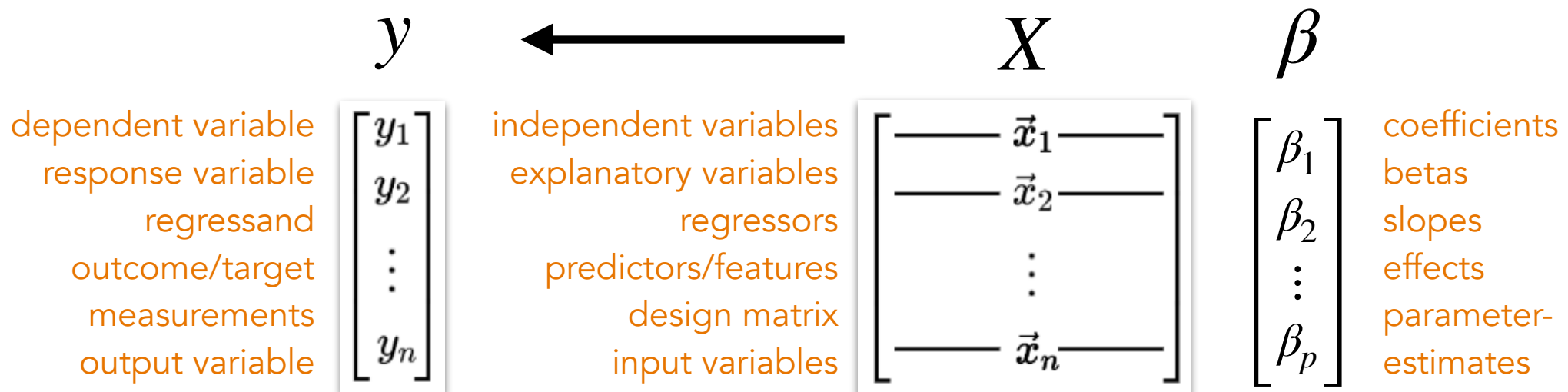
coefficients
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$$\begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}$$

The General Linear Model (GLM)

how do we learn this? $y = X\beta + \epsilon$

Models a **scalar** variable **y** as a linear function of predictor **variables X**

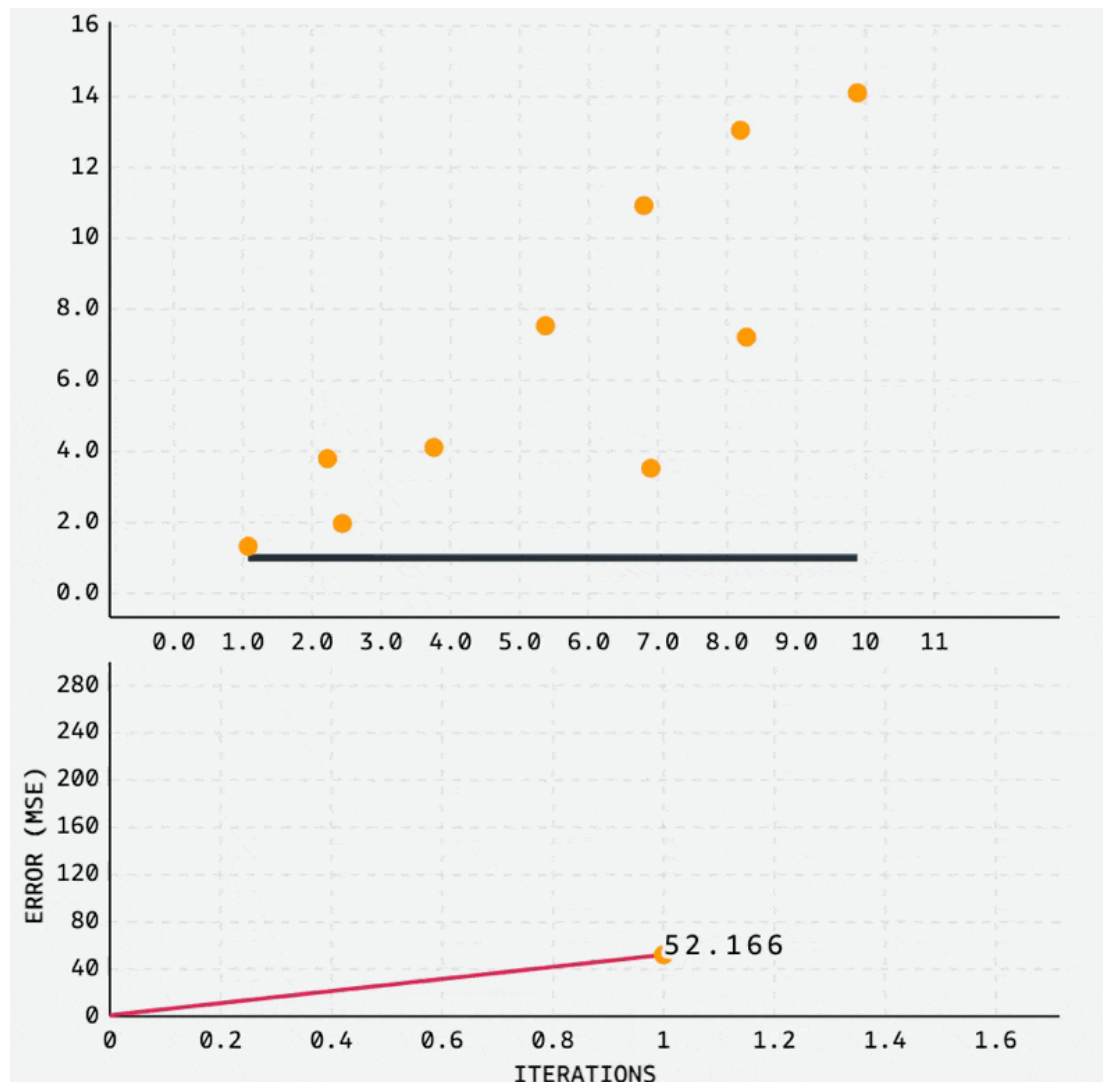


Approach One: Iterative Learning

("thoughtful guess-timation")

Simple (univariate) regression

$$Y_i = \beta_0 + \beta_1 X_{1i} + \epsilon_i$$



Multiple regression

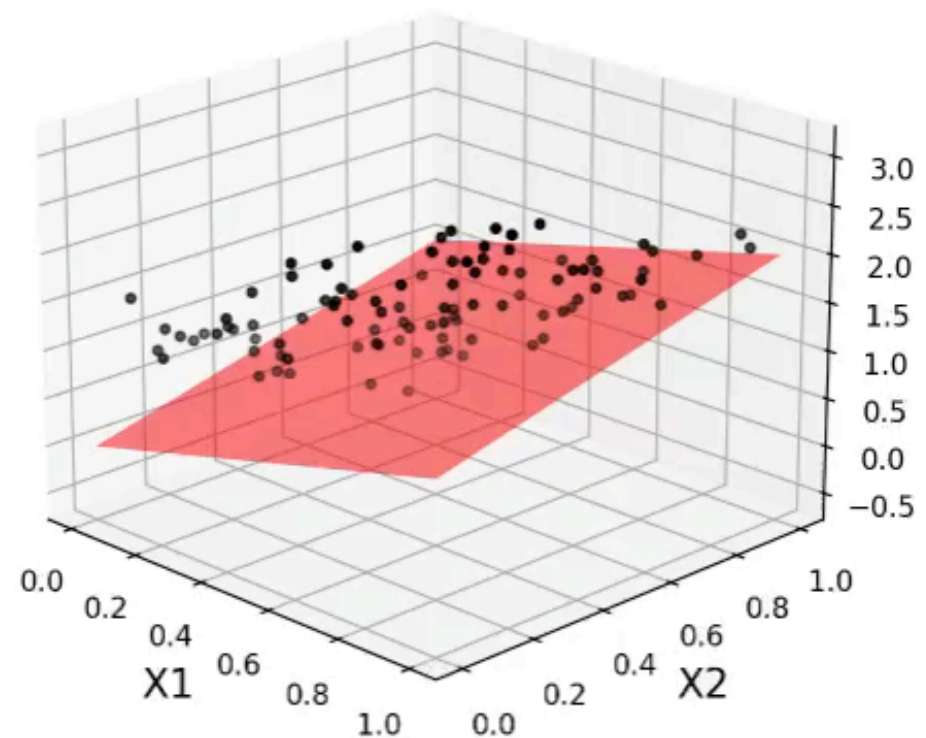
$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

Intercept:

Slope1:

Slope2:

Iterative fitting (MSE: 0.9420)



Approach Two: Ordinary-Least-Squares

$$y = X\beta + \epsilon$$

Ordinary Least Squares (**OLS**):

an *analytic* (closed-form) equation for estimating betas that minimizes SSE!

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$\hat{\mu}$ Mean = best **single estimator** that minimizes SSE

$\hat{\beta}$ OLS = best **multi estimator** that minimizes SSE

What is it doing?

Calculate the **similarity** between **each X and y**
after removing the **similarity between Xs**

What is *unique contribution* of each x in predicting y?

undo

Matrix Inverse: think **division**

$$x * \frac{1}{5} = \frac{x}{5}$$

Matrix-Matrix dot-product:
like *looping* over all possible
pairs of columns in X

Matrix-Vector dot-product:
like *looping* over each single
column of X

Similarity between
predictors

$$\hat{\beta} = \underbrace{(X^T X)^{-1}}_{\text{Similarity between predictors}} \underbrace{X^T y}_{\text{Similarity between predictors and outcome}}$$

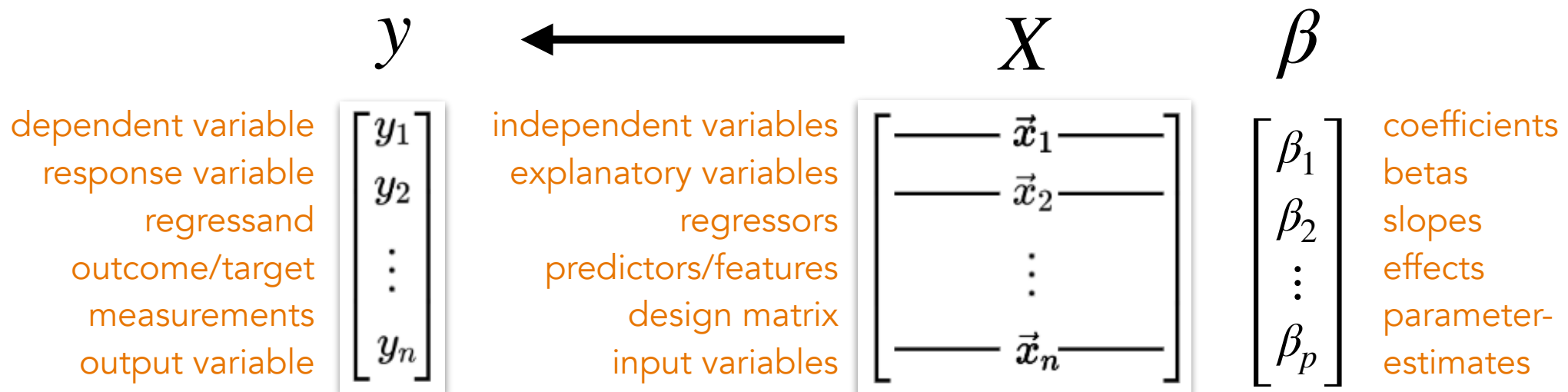
Similarity between
predictors and outcome

Design Matrix: The “eyes” of the GLM

Design Matrix

$$y = X\beta + \epsilon$$

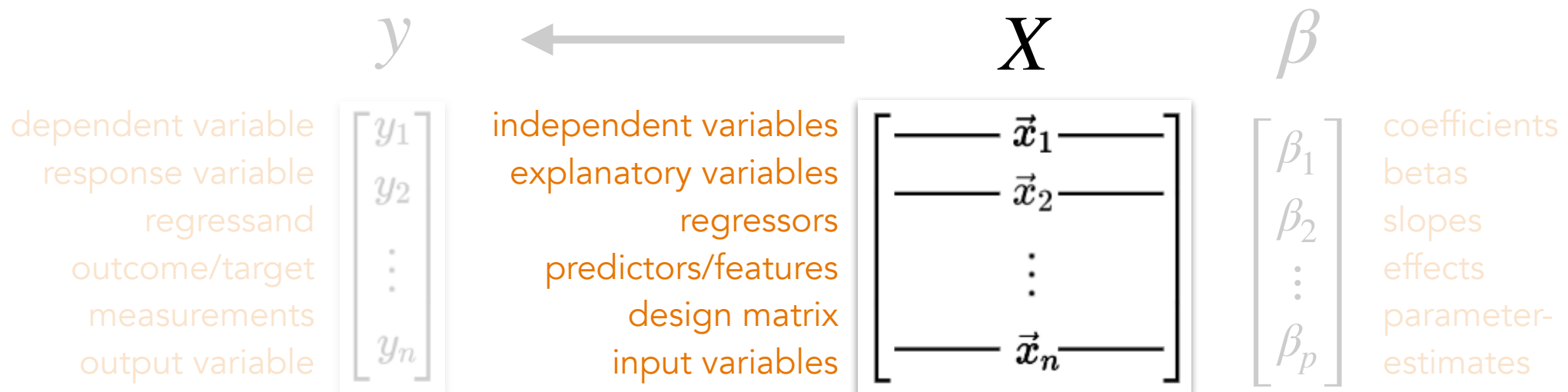
Models a **scalar** variable **y** as a linear function of predictor **variables X**



Design Matrix

$$y = X\beta + \epsilon$$

Models a scalar variable y as a linear function of **predictor variables X**



New Dataset: Credit-Card Debt

index	income	limit	rating	cards	age	education	gender	student	married	ethnicity	balance
1	14.89	3606	283	2	34	11	Male	No	Yes	Caucasian	333
2	106.03	6645	483	3	82	15	Female	Yes	Yes	Asian	903
3	104.59	7075	514	4	71	11	Male	No	No	Asian	580
4	148.92	9504	681	3	36	11	Female	No	No	Asian	964
5	55.88	4897	357	2	68	16	Male	No	Yes	Caucasian	331
6	80.18	8047	569	4	77	10	Male	No	No	Caucasian	1151
7	21.00	3388	259	2	37	12	Female	No	No	African American	203
8	71.41	7114	512	2	87	9	Male	No	No	Asian	872
9	15.12	3300	266	5	66	13	Female	No	No	Caucasian	279
10	71.06	6819	491	3	41	19	Female	Yes	Yes	African American	1350

variable	description
income	in thousand dollars
limit	credit limit
rating	credit rating
cards	number of credit cards
age	in years
education	years of education
gender	male or female
student	student or not
married	married or not
ethnicity	African American, Asian, Caucasian
balance	average credit card debt in dollars

But this isn't a number...



Do **students** have a different **credit card balance** from **non-students**?

How does the GLM **see** different variables?

index	income	limit	rating	cards	age	education	gender	student	married	ethnicity	balance
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But this isn't a number...

Do **students** have a different **credit card balance** from **non-students**?

How does the GLM **see** different variables?

My favorite metaphor:



Continuous Variables



Categorical Variables

How does the GLM **see** different variables?

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More than 2 settings

How does the GLM **see** different variables?

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**Many
simultaneous
continuous
variables**



**Many
simultaneous
categorical
variables**

How does the GLM **see** different variables?

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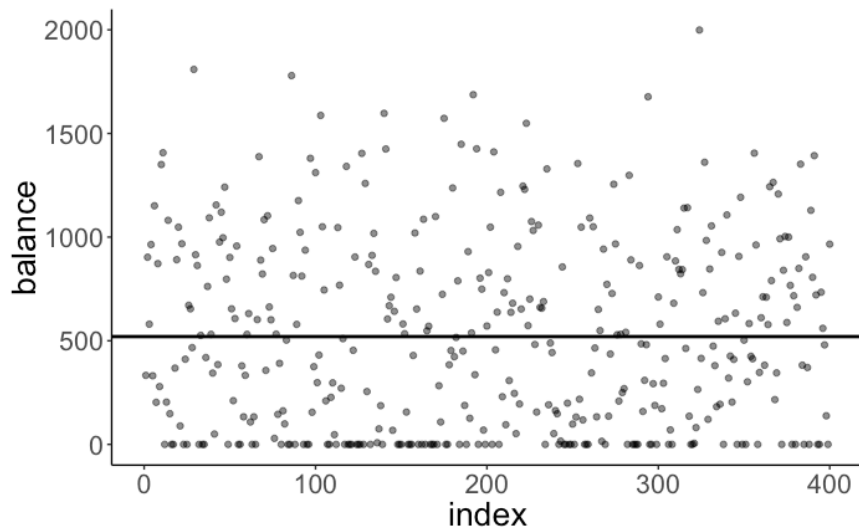
Do **students** have a different **credit card balance** from **non-students**?

H₀: Students and non-students have the same balance.

Compact Model

$$\hat{Y} = \hat{\beta}_0 + \epsilon$$

Model prediction



same $\hat{Y}$ regardless of student/not

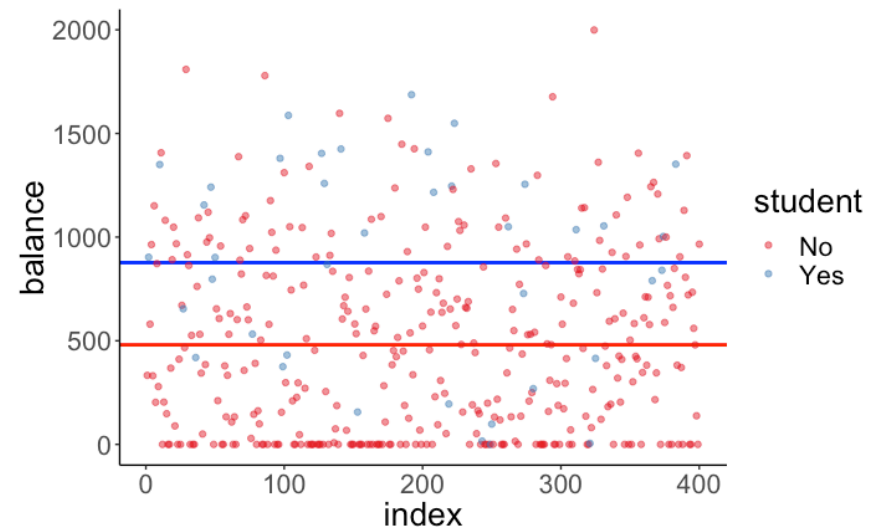
H₁: Students and non-students have different balances.

Augmented Model

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X + \epsilon$$

Model prediction

student (yes/no)



switches $\hat{Y}$ if student/not

Design Matrix: The “eyes” of the GLM

- Categorical Variables
 - We can **encode** them into numbers using a variety of approaches
 - A variable with **k unique levels** is represented with **k-1 columns** in the Design Matrix

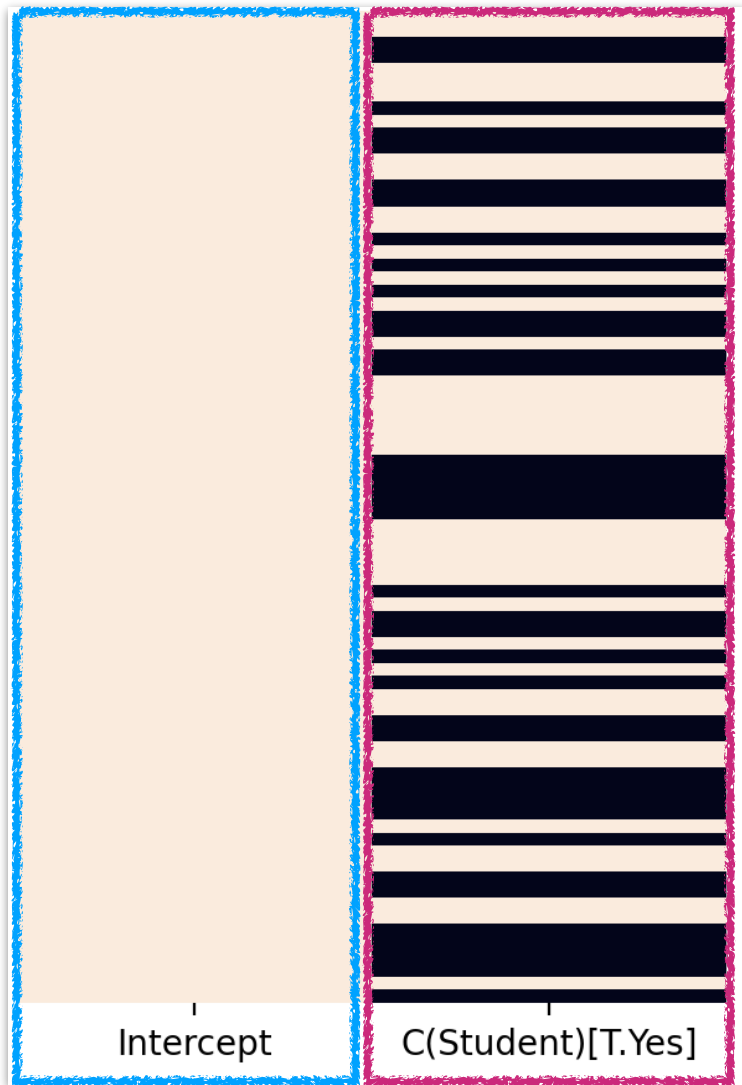
Balance	Student
i64	str
16	"Yes"
1216	"Yes"
148	"No"
108	"No"
532	"Yes"

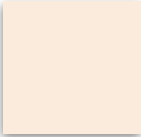
Yes = 1

No = 0

```
array([[1., 1.],  
       [1., 1.],  
       [1., 0.],  
       [1., 0.],  
       [1., 1.]])
```

Design Matrix: The “eyes” of the GLM



 = 1

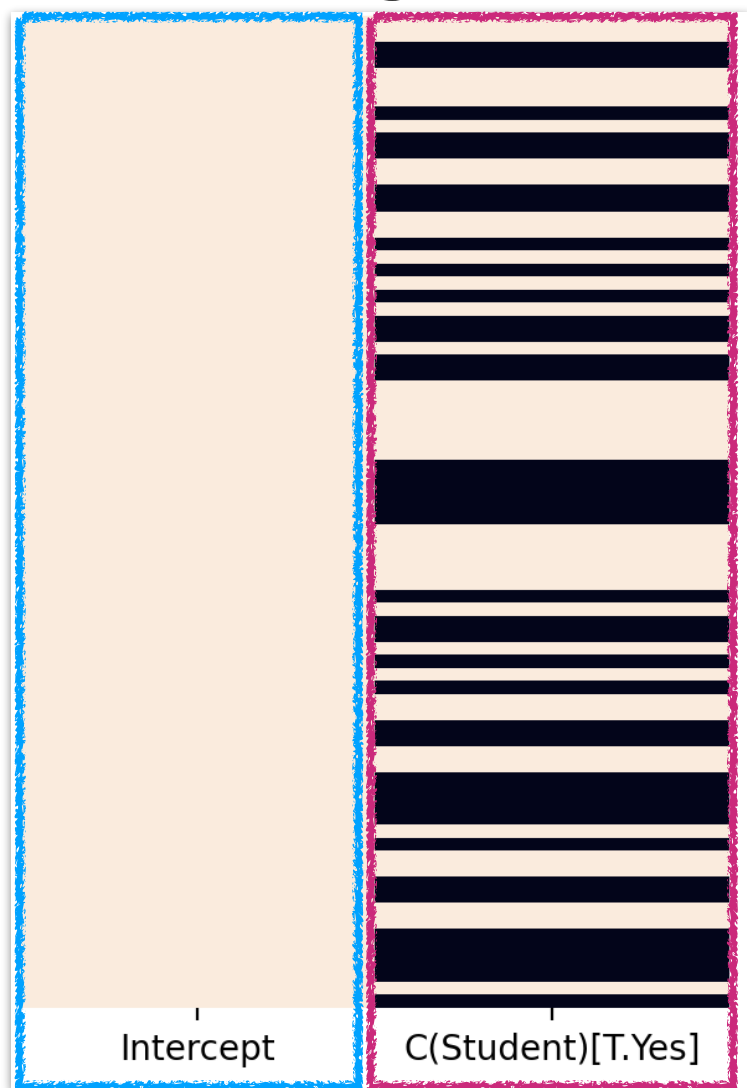
 = 0

(mean of No)(Yes - No)

Balance	Student
i64	str
16	"Yes"
1216	"Yes"
148	"No"
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532	"Yes"

```
[[1., 1.],  
 [1., 1.],  
 [1., 0.],  
 [1., 0.],  
 [1., 1.]]
```

Encoding 1: Treatment (Dummy) Codes



 = 1
 = 0

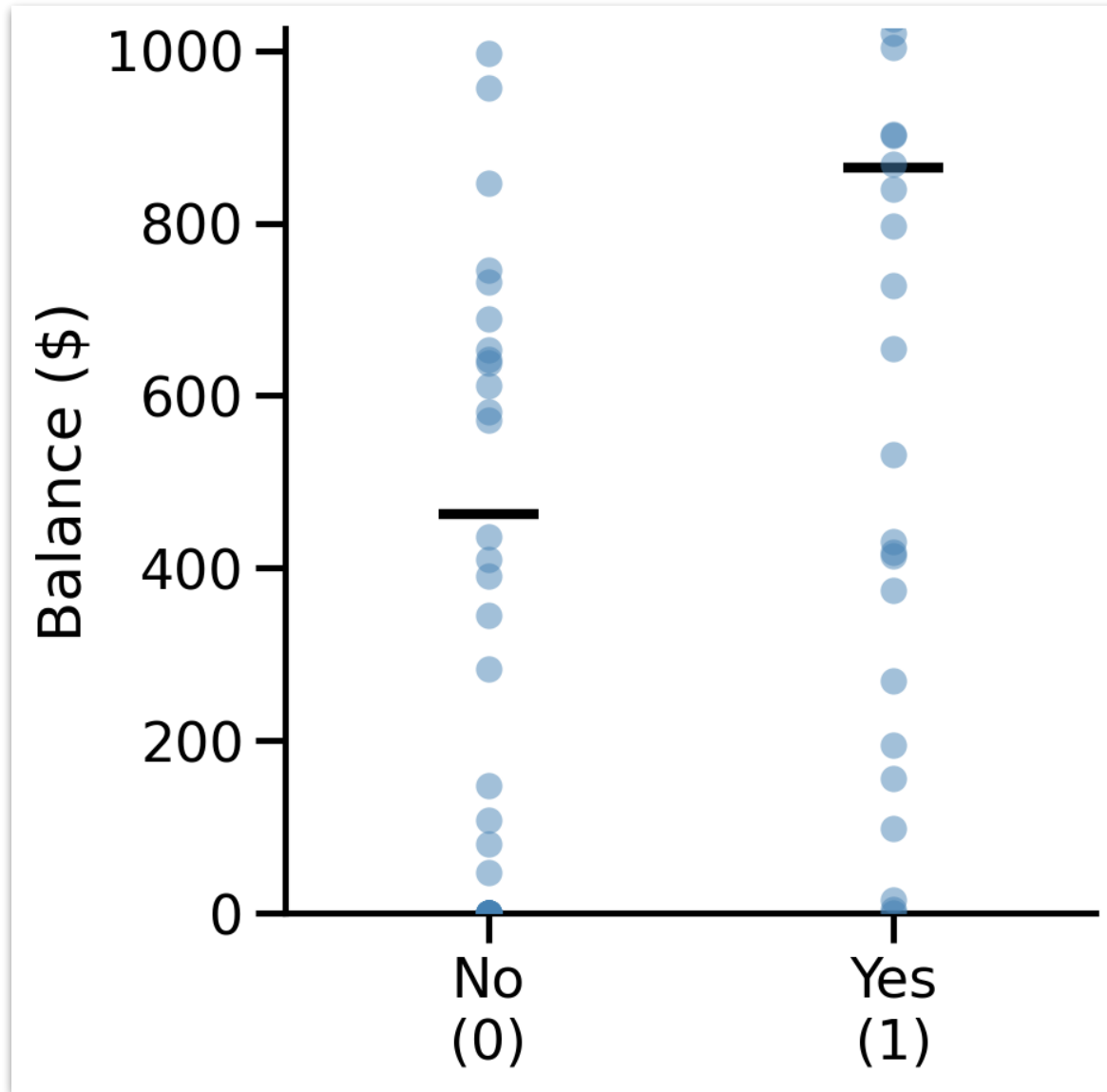
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```
[[1., 1.],  
 [1., 1.],  
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 [1., 0.],  
 [1., 1.]]
```

- **Reference level** is coded as 0, other level(s) coded as 1
- **Intercept** = reference-level (mean)
- **Slope(s)** = difference from reference-level mean

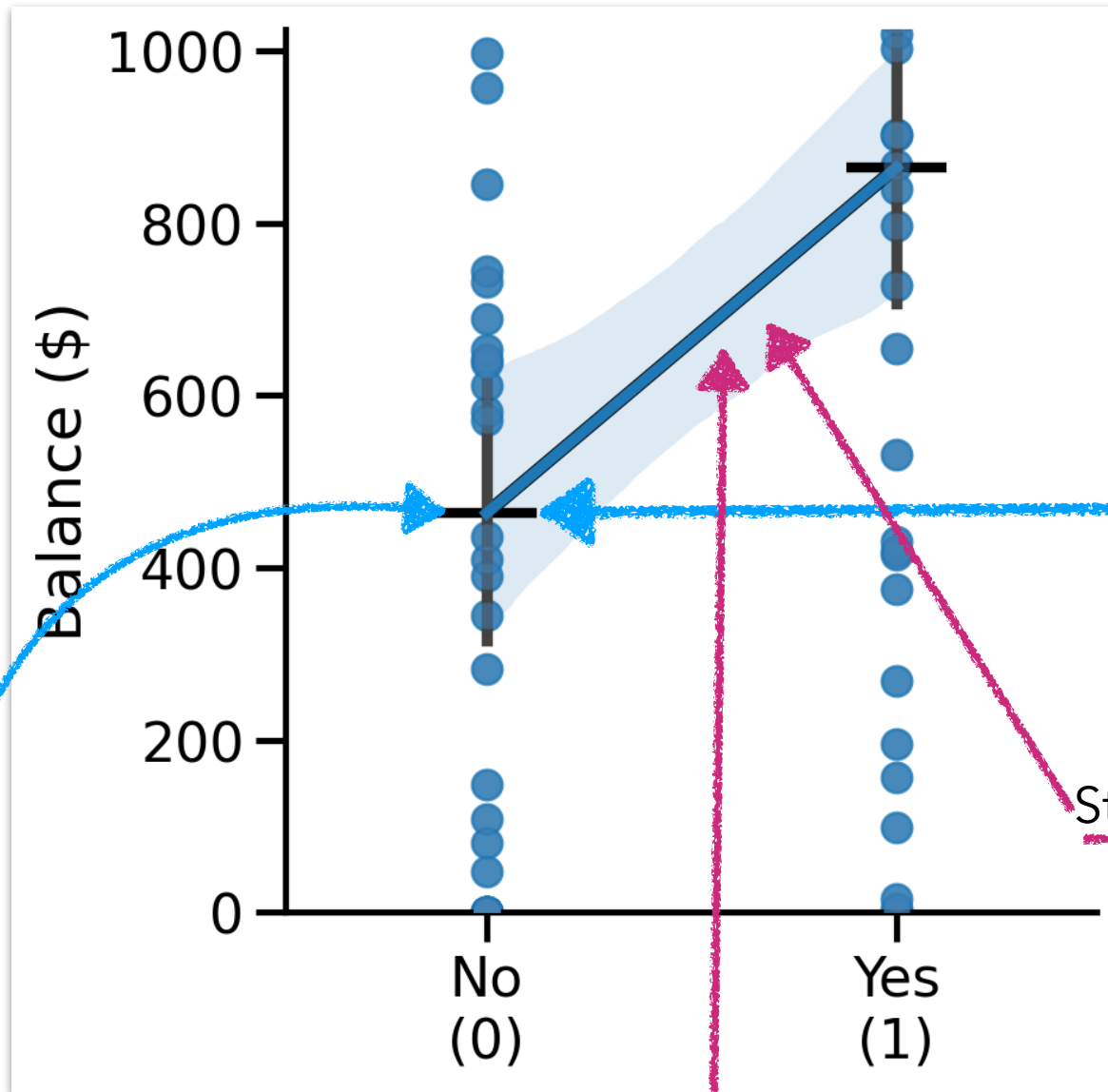
Treatment (Dummy) Coding Visually



0: Student = No
1: Student = Yes

- **Reference level** is coded as 0, other level(s) coded as 1
- **Intercept** = reference-level (mean)
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Treatment (Dummy) Coding Visually

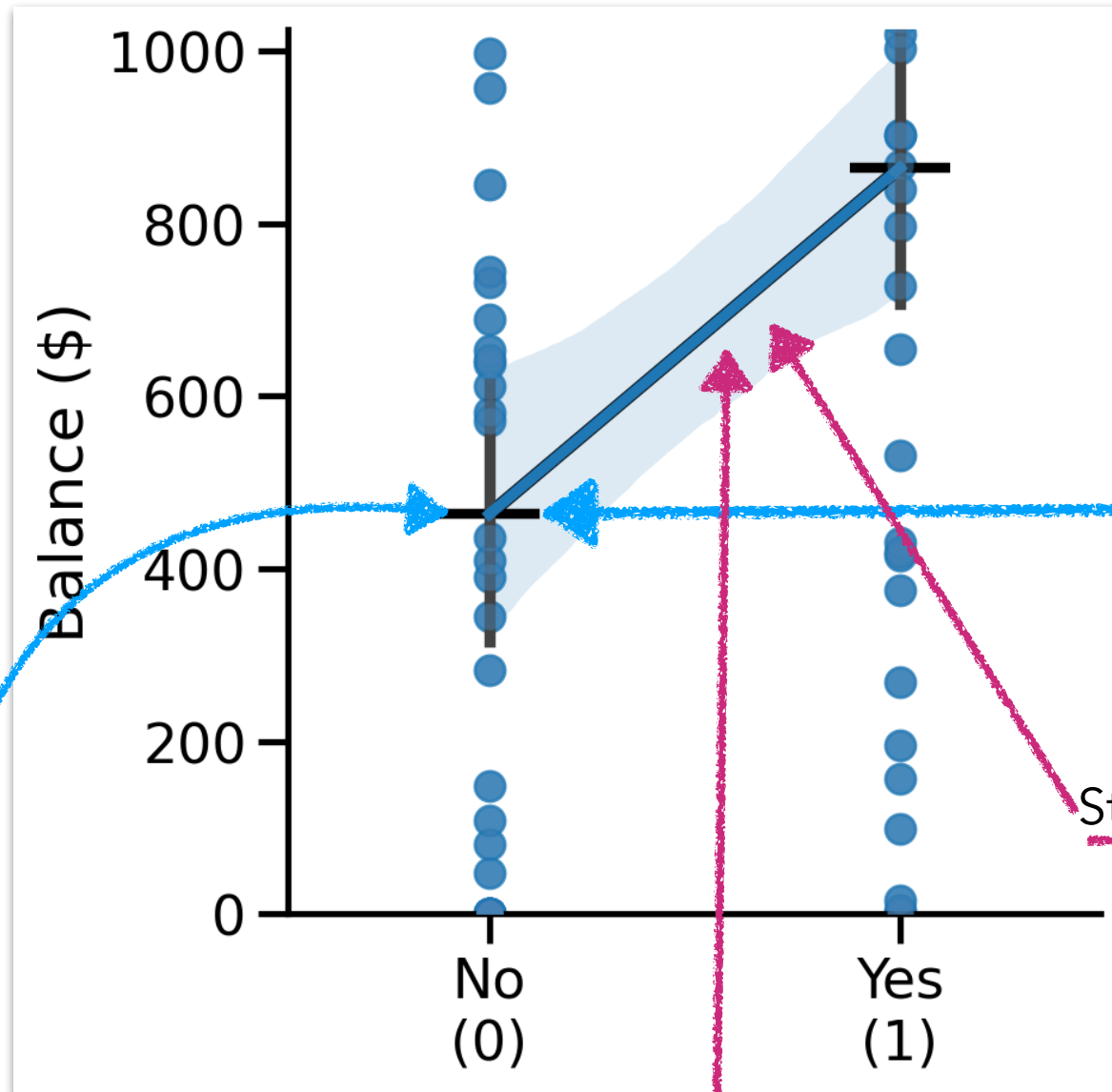


0: Student = No
1: Student = Yes

Student (Yes) - Student (No)

- **Reference level** is coded as 0, other level(s) coded as 1
- Intercept = reference-level (mean)
- **Slope(s)** = difference from reference-level mean

Linear Model (GLM) of Mean Differences

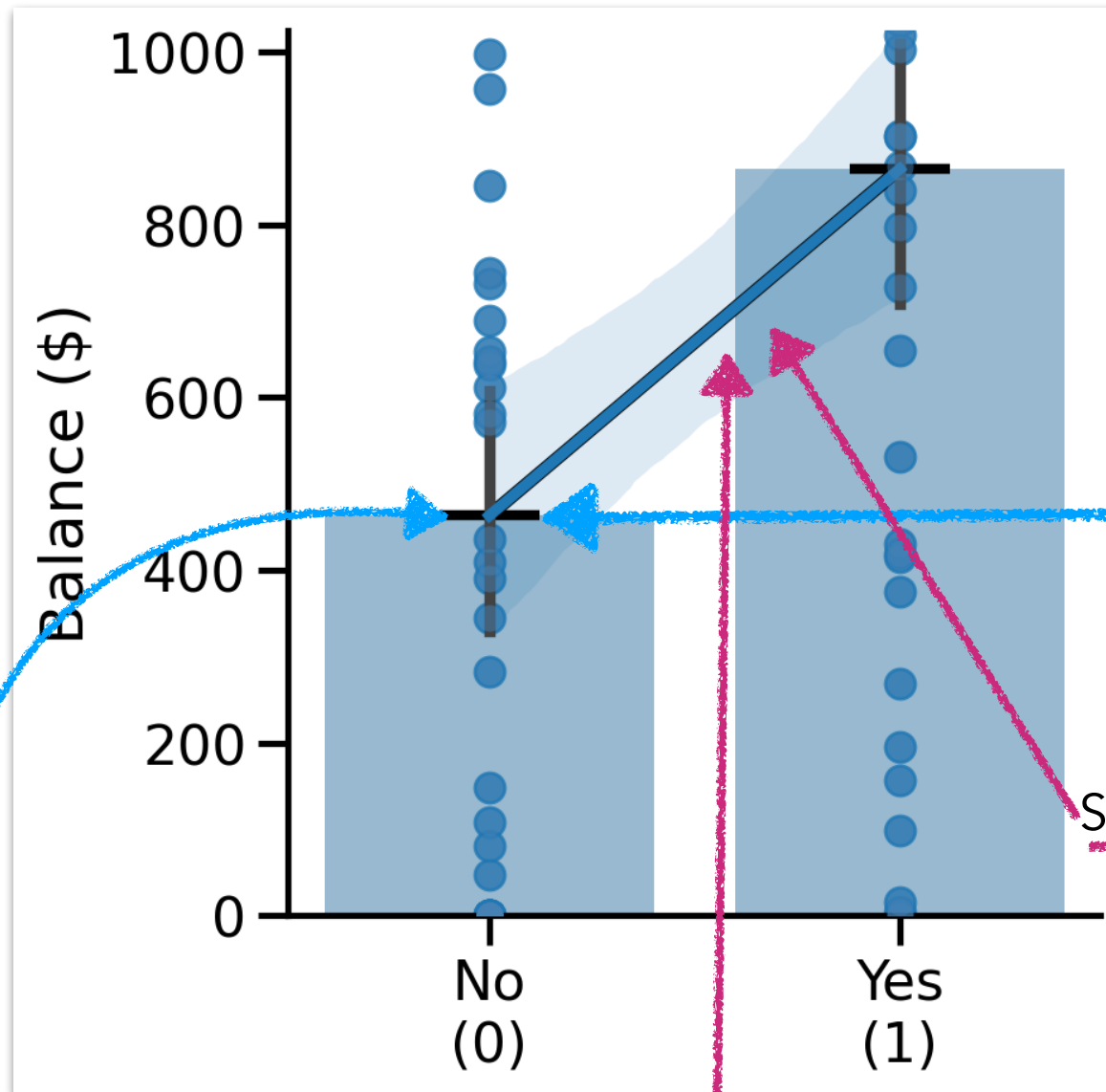


0: Student = No
1: Student = Yes

Student (Yes) - Student (No)

- **Reference level** is coded as 0, other level(s) coded as 1
- Intercept = reference-level (mean)
- **Slope(s)** = difference from reference-level mean

Independent Samples T-test



0: Student = No
1: Student = Yes

Student (Yes) - Student (No)

- **Reference level** is coded as 0, other level(s) coded as 1
- Intercept = reference-level (mean)
- **Slope(s)** = difference from reference-level mean

Design Matrix: The “eyes” of the GLM

$$y = X\beta + \epsilon$$

- **Mixing-board metaphor**
 - Continuous valued columns are like sliders
 - Categorical valued columns are like switches
- **Categorical Variables**
 - We can **encode** them into numbers using a variety of approaches
 - A variable with **k unique levels** is represented with **k-1 columns** in the Design Matrix

For the remaining time...

**Get a jump-start on the Linear algebra
videos on the course website!**